Dealing with Mold in Damaged Homes

The longer mold grows, the greater the potential hazard and the harder it is to control.

by Claudette H. Reichel

Hurricane Katrina caused unprecedented damage in the United States. Louisiana residents have recovered from many floods and hurricanes in the past, but never before have so many been hit so hard that they were unable to return to their homes for so long. Never before has such a large American population experienced such an extended flood period and power outage combined with high summer humidity. And never before have so many residents returned not only to homes that were torn up physically by the storm, but also to homes that hosted an unprecedented population explosion of mold.

Molds are a category of fungi. They serve as nature's recycler by helping to break down dead materials. Molds produce tiny spores that float and spread easily through the air, so they are just about everywhere. Live spores act like seeds, forming new mold growths (colonies) when they find the right conditions—moisture, nutrients (nearly anything organic), and a suitable place to grow. Of these, moisture in either liquid or vapor form is the key factor—both for growth and for control.

Mold begins to grow on materials that stay wet longer than two or three days. The longer it grows, the greater the potential health hazard and the harder it is to control. So as soon as it is safe to return, it's crucial for the homeowner to begin cleanup and to dry out the house as safely and effectively as possible. After such a massive disaster, many if not most homeowners faced the task of doing this on their own, with limited help and resources.

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Heavy mold tracks the obvious flood level on this kitchen's paper-faced drywall. Note that the inspector is wearing an N-95 respirator.

First Things First

If your home has been flooded, take photographs before cleaning up for insurance purposes. Today, most homeowners' insurance policies do not cover mold damage or cleanup costs, but flood insurance may cover these expenses.

Once you have taken the photographs, you should get started as soon as possible. Don't wait for the claims adjuster to see your home before you remove wet and moldy materials.

If you are hiring a contractor to clean your home, seek out licensed or certified mold remediation contractors with special training and equipment. Get in writing the cost, methods, and steps to be used. Talk to several contractors and compare each contractor's procedures with the following guidelines. Effective procedures may vary, depending on what specialized equipment and chemical treatments the contractor uses, but the contractor that you choose should address the principles listed in the following guidelines.

Mold and Health

People are exposed to mold mainly by breathing spores or other tiny organic fragments. People can also be exposed through skin contact—for example, by touching moldy surfaces—and by eating mold-contaminated food. The health effects of exposure vary widely and are usually hard to predict. They depend on the sensitivity of the person, the amount and type of exposure, the length of exposure, the type of mold, and many other factors.

Some people are much more sensitive to mold than others, but long-term or heavy exposure is unhealthy for anyone. Mold commonly triggers allergic reactions and asthma attacks, may lower resistance to illness, and may affect people's health in other ways. Young children, the elderly, and the ill are most vulnerable.

Many types of mold can produce harmful chemical compounds called mycotoxins, but they can do so only under certain conditions. If a toxin is produced, it may be present in both live and dead spores and in fragments in the air. The commonly used term "black mold," when used to refer to a species that produces toxins, is misleading, since many types of mold are black. Although the potential damaging effects of specific mycotoxins are known and varied, identifying a mold that can produce mycotoxins does not tell you whether you have been exposed to a toxin. Still, all indoor mold growth is potentially harmful and should be removed promptly, no matter what type of mold is present, and no matter whether it can produce a toxin.

Mold testing to identify the type or quantity of mold in a house is not usually necessary, and it cannot tell you whether you have been exposed to a toxin, as I have just explained. Some insurance companies and legal services may require sampling for evidence. Professional mold remediation contractors may test before and after cleanup to verify that the mold is no longer present.

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This New Orleans home was flooded for weeks, then sat wet for months before the homeowner was allowed to return and face the results.

Do It Yourself: Ten Steps to Cleaning Up Mold

To clean up safely and effectively, follow these steps and refer to one of the following publications. *A Brief Guide to Mold, Moisture, and Your Home* and *Mold Remediation in Schools and Commercial Buildings*, both published by EPA, are available online at www.epa.gov/mold. The CDC's *Mold Prevention Strategies and Possible Health Effects in the Aftermath of Hurricanes Katrina and Rita* is also available online at www.bt.cdc.gov/disasters/mold/report/.

Wear protective gear during cleanup. Wear gloves, goggles, rubber boots, and a particle respirator that has received a National Institute for Occupational Safety and Health (NIOSH) rating of N-95 or higher. Some types of respirator have valves to make it easier to breathe. For heavy, long-term exposure, a properly fitted half-face or full-face respirator is recommended.

Isolate work area and ventilate to the outdoors. Disturbing mold colonies during cleanup can cause a huge release of spores into the air, so seal off the moldy areas from the rest of the house. Open windows, and don't run the central air system during cleanup. Tape plastic over air grilles, and close off the stairwell with taped plastic if the second story is dry and clean. If the power is on, put a fan in a window, facing out, to exhaust mold-filled air to the outdoors.

Remove moldy porous materials. Porous moldy or sewage-contaminated materials should be removed, put in plastic bags if possible, and thrown away. To reduce the release and spread of mold spores, it is helpful to cover moldy material with plastic sheeting before removing it.

Remove all flooded carpeting, upholstery, fabrics, and mattresses right away. It's best to discard them, but if you hope to salvage a valuable item, have it cleaned, disinfected, and dried quickly outside the home. Never reuse flooded carpet padding.

Remove all wet fibrous insulation—even if the wallboard appears to be dry. Wet insulation will stay wet for far too long, leading to the growth of unhealthy mold and decay fungi inside the walls. Cut wall covering above the level that was wet; water can wick up above the flood level—sometimes as much as 3–4 feet above it.

It's best to remove all moldy, porous materials, especially if there is heavy or long-term mold growth—including gypsum wallboard, processed wood products, ceiling tiles, and paper products. Plaster, wood paneling, and non-paper-faced gypsum board walls that have dried, are in good condition, and are not insulated can be cleaned and sanitized to salvage them. There is a risk of mold on the back side, however, and this mold can release spores into the home through air leaks in the walls. Remove vinyl wallpaper, flooring, and other coverings on wet materials.

First clean, then disinfect. Surface mold can be effectively cleaned from nonporous materials such as hard plastic, concrete, glass, and metal; solid wood can also be cleaned, since mold grows only

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Different bands of mold mark these walls. Do they indicate multiple flood levels or wicked moisture?

on its surface. Cleaning should remove, not just kill, the mold, because dead mold spores and fragments can cause allergic reactions, and some may still contain harmful toxins. Wash dirty or moldy materials with nonphosphate all-purpose cleaners, because phosphate residue is mold food. Rough surfaces may need to be scrubbed. Rinse, but don't use a high-pressure spray that can force water into materials. Be sure to remove any sediment. Hose out opened wall cavities.

After cleaning, you may use a disinfectant product to kill any mold missed by the cleaning. Don't try to disinfect until after you have cleaned, as soil can make some disinfectants, such as bleach, less effective. If there was sewage contamination, you must disinfect. If you disinfect, follow the directions and warnings on the label, and handle all disinfectants carefully; never mix bleach with ammonia or acids. Many disinfectants kill molds, but do not prevent the growth of new mold if the material remains damp.

On colorfast, nonmetal surfaces, you can disinfect with a solution of $\frac{1}{2}$ to 1 cup of household chlorine bleach to 1 gallon of water. Do not use a bleach solution in the air conditioning system. Milder, less corrosive disinfectants include alcohols, phenolics, hydrogen peroxide, and other commercial disinfectants.

Consider borate treatment. Applying a borate treatment to wood framing can provide multiple benefits; it allows the wood to resist termites, decay, and mold. Treatments formulated to penetrate wood offer greater protection. Fungicides can also help to inhibit the regrowth of mold during drying. Do *not* apply sealants to wood or interior surfaces. If desired for added protection from any remaining mold, cleaned surfaces can be encapsulated with latex paint or other permeable coatings that do not form a water vapor barrier.

Flush the air. After cleaning and disinfecting, air out the building. Open all windows and use fans in windows on one side of the house to pull mold spores to the outdoors. Be careful not to depressurize the house to the point where you are pulling contaminated air from the crawlspace into the house.

Speed dry. After cleaning and disinfecting, dry all materials as quickly as possible. In warm weather, close all the windows and run the air conditioner to remove moisture from indoor air (run the air conditioner on auto, not constant fan). In cool weather, heat the home to reduce the relative humidity. Run fans and use a dehumidifier if possible. If there is no power, keep the windows open.

Remain on mold alert. Continue looking for signs of moisture or new mold growth. New mold can form in as little as two days if materials stay wet. Wood and other materials that may look dry can still be wet enough to support regrowth. If mold returns, repeat cleaning and if possible use professional speed-drying equipment and moisture meters. Regrowth may signal that the material was not dry enough, or that it should be removed.

Do not restore until all materials have dried completely. Wood moisture content should be less than 20% before you restore the interior. Do *not* use vinyl wallpaper, oil-based paint, or other interior finishes that block drying to the inside.

Restore with flood-resistant materials. Consider restoration with "wet floodproofing" in mind to reduce damage from future flooding. Choose water-resistant floorings such as tile, with area rugs that can be removed before the storm. Consider installing drainable, cleanable walls with rigid or spray foam insulation, removable wainscoting, or paperless drywall.

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For more information:

To learn more about storm cleanup, see The AgCenter publications *Storm Recovery Guide for Homeowners*, and *Cleaning Your Flood Damage Home Fact Sheet*, both available from www.lsuagcenter.com.

To learn more about cleaning up mold, see *A Brief Guide to Mold, Moisture, and Your Home*, and *Mold Remediation in Schools and Commercial Buildings*, both available online at www.epa.gov/mold; and *Mold Prevention Strategies and Possible Health Effects in the Aftermath of Hurricanes Katrina and Rita*, available online at www.bt.cdc.gov/disasters/mold/report/.